

Where do
10,00,000+
electronics
design
enthusiasts
meet every
month?



/DesignElectronics

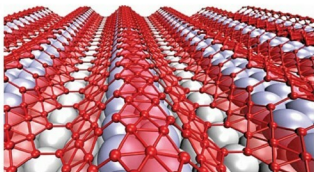


What about you?

World's leading social media
platform for electronics design
professionals & enthusiasts!

facebook.com/DesignElectronics

MATERIALS



Wavy borophene (Image courtesy: Zhuhua Zhang/Rice University)

graphene and the polymers currently used. Superconductivity of borophene is critical. Borophene grown on a silver substrate takes a wavy form, which makes it suitable for flexible electronics. As a 2D metal, borophene helps fill a void in the family of 2D nanoelectronic materials.

With the synthesis and characterisation of borophene researchers have moved a step ahead. Properties such as low mass density, high strength and high levels of flexibility make borophene a promising candidate as a reinforced element for designing composites and for flexible nanodevices, such as flexible electrodes and contacts for nanoelectronics.

Development status

Most of the work with borophene so far has been aimed at synthesis as well as characterisation of the material's properties. Advanced computer simulation methods showed formation of crinkles in the corrugated surface.

Borophene is a proposed allotrope of boron made up of single-atom sheets. The research reveals that one cluster consists of 36 boron atoms arranged in a two-dimensional sheet with a perfect hexagonal hole in the middle, which is of real significance. It suggests that theoretical calculations about a boron planar structure might be right.

Although bulk allotropes of carbon and boron differ greatly, in nano regime small clusters of these elements show remarkable similarities. Boron analogues of two-dimensional carbon allotropes such as graphene have been predicted. Researchers demonstrated theoretically that borophene is enriched with more interesting electronic characteristics than graphene but a strong foothold in production was missing.

Theoretical studies predict that 2D boron sheets will adopt an atomic configuration similar to that of boron atomic clusters. Recently, researchers successfully synthesised borophene using molecular beam epitaxy. They reported the formation of two-dimensional boron by depositing the elemental boron onto a silver surface under ultra-high-vacuum conditions. The graphene-like structure was buckled, weakly bonded to the substrate, and metallic.